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1. A ~~X~~

2. C ✓

3. A ✓

4. A ~~X~~

5. D ~~X~~

6. C ~~X~~

7. A ✓

8. C ✓

9. A ✓

10. B ✓

11. B ✓

12. C ✓

13. B ✓

14. C ✓

15. A ~~X~~

16. C ✓

17. A ✓

18. C ✓

19. B ✓

20. C ✓

21. Supervised learning is a type of ML algorithms that uses labeled dataset to train the model with a specific target.

22. The model M_1 is overfitting because it has a very high accuracy when training, but a very low accuracy when testing.

23. Regression analysis is a process of analysing the predicted value of a regression model.

24. Two challenges of Machine learning:

- Gather the data and make sure the data is legit, accurate and useful
- Choosing the correct model for the dataset

25. Two major application of data science:

- Business - To analyze the trends, and satisfaction rate of customer
- Banking: Analyze the customer before giving the loan

26. Distinguish structured and semi-structured data:

- Semi-structured data: The data that has some sort of hieritcal. For example = sql
- Structured data: The data that has a very high-level of hieritcal and be understood by machine. For example: html tags,

27. We can find outlier in the data by plotting the data and verify or by using the formula $Q_1 - 1.5IQR$ ~~for~~ and $Q_3 + 1.5IQR$. If the ~~output is lower~~ data is lower than $Q_1 - 1.5IQR$ it will be an outlier under the group range, and if the number is more than $Q_3 + 1.5IQR$ it will be an outlier over the group range.

28. Data-preprocessing is a process of modifying the data before fitting it into the model. This process often involve data cleaning, missing value treatment, outlier detection, data normalization etc.

29. Data normalization is required in K-NN in order to scale the big data into smaller range so the distant calculation would be faster.

30. Continuous attributes are discretized by using data discretization method such as putting the data in Bin and apply operation such as replacing with min/max value.

31. K in K-mean algorithm refer to the number of neighbor that will be used for label voting. For example, if $K=3$, mean that the 3 nearest neighbor will be use for label voting

32. Hyper parameters of ML algorithms:

- Random Forest

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33. Gradient descent is



34.

- SVM1 = This is the best because the linear classification is correct and the space between the line is wide

- SVM2 = This is a bad svm because it classify incorrectly

- SVM3 = This is a bad svm because it classify correctly but the space between line is very small.

35. Radical Basis Function kernel work by changing the dimension of a dataset so hyperplane and classification can be applied correctly.

